## Ex1 - Visual Navigation for Drones

In this assignment, we focus on GNSS-DENAYED navigation, in particular, Visual Navigation:

Visual Navigation (for drones) focuses on the following problem:
Given a video from a drone, detect the drone position (without the use of GNSS). In particular, the following interesting sub-problems can be considered:

1. Given a video feed from a drone. Implement a semantic segmentation of common classes such as humans, cars, buildings, roads, roundabouts, trees, antennas, and pedestrian crossings.
2. Given a video of the drone, allow it to remain in the same position (more or less) (aka hover).
3. Given a video of a drone from its launch point to a given time, allow the drone to “return to Home” (aka RTH) based on the video feed.
4. Given a predefined map (some kind of a GIS DB such as Google Earth), design an algorithm that allows the drone to fly to a given point on the map.

**In this assignment, we mainly focus on the following optical navigation problem:**
Given data from a drone flight (video + telemetry, including GNSS positioning, barometric height, camera angle - see SRT files below). Preprocess this flight data, such that given a new real-time flight data (Video stream and telemetry - without GNSS), the position at which the drone is looking on can be computed in real-time (the coordinate of the center point of the video, assuming that the preprocessing data contains the current position).

**Data (Videos & Images):**

1. DJI mini3 pro: Use the following videos: v11, v12, v13, v14. The corresponding telemetry files can be found here. The videos were shot by a DJI mini 3 pro, at 1080p @ 30 fps, at a 60-degree angle @ a height of 119 meters (above the takeoff point). Use the additional two low-flying video1 and video2,
2. DJI Air 2s (100 m): v1, v2
3. Autel Robotics nano plus (120 m): v3, v4
4. Night time video: NV1, NV2
5. Hi Fly (500 meters): V1, V2,
6. DJI Air 3 (120 meter @ 45 deg): v1, (50 meter @ 45 deg): v2. The original files are in this folder - including the SRT files (can be seen via VLC, or using this repo).

**Directions:**

1. Write a literature review on algorithms and tools for visual navigation (for low-flying drones 20-200 meters). Do focus on recent research papers based on open-source implementation (aka “paper with code”).
2. Design a complete algorithm for both the preprocessing stage and the navigation stage.
3. Search for the most suitable platform for drone optical navigation and edit it so it will work on the suggested videos, as defined in the problem of interest.

4. Perform a preliminary experiment, in which you use a video (from the testing folder - TBD) and, given the angle of the camera, it computes the expected path of the drone. This path will be compared with the captured path (as in the SRT file).

**Remarks:**

1. You are mostly welcome to use any LLM tool you want, yet make sure you deeply understand the underlying methods and algorithms.
2. There are many GitHub repositories that you are strongly recommended to use as a reference.
3. Your solution should be a GitHub repo with fully documented algorithms. An easy-to-build code.
4. Your Solution will be presented in class!